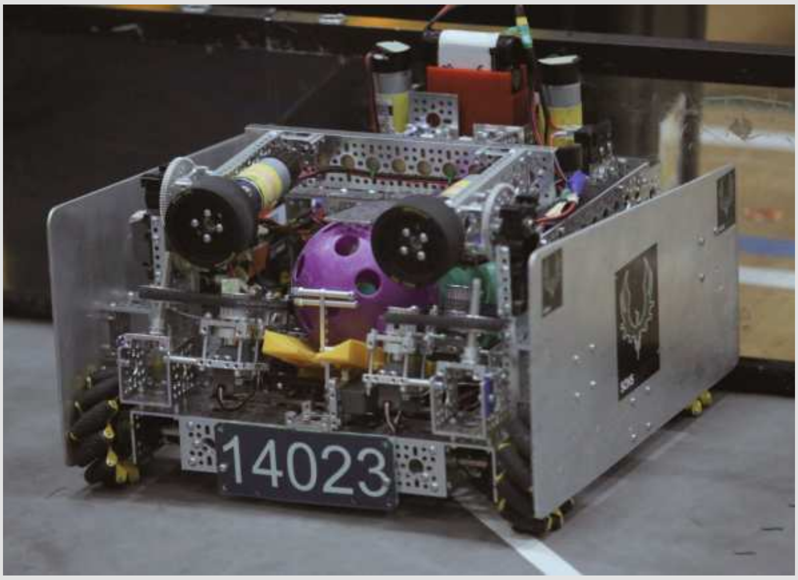
2025-2026

ENGINEERING PORTFOLIO



14023

SPRINGBANK ROBOTICS TEAM

PHOENIX RESURRECTION

THE TEAM


Liam
Team Lead
Build/Design


Charles
Team Lead
Coach/Design


Hunter
Team Lead
Drive/Build


Ethan
Main Drive
Build/Strategy


Laurana
Code/Build
Design


Arianna
Code
Portfolio


Riley
Code


Fiona
Junior Drive
Strategy

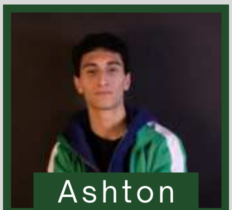

Carter
Junior Drive
Strategy


Simon
Junior Build


Keiran
Code


Eliana
Junior Build
Strategy

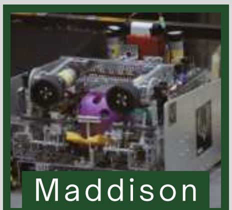

Angil
Junior Build


Ashton
Junior Code
Strategy


Charlie
Junior Code


Harrison
Junior Build


Phoenix
Mascot


Maddison
Robot


Mr. Mathew
Mentor


Ms. Heidi
Mentor

We are **Springbank Robotics Team (SRT) Phoenix Resurrection, 14023** from Calgary, Alberta. We're a student-led, veteran team in our 9th year in FTC.

Our robotics team is lucky to have mentors who **guide us, give us feedback, and support our ideas.** However, at the end of the day, we are a **student-led team, where the kids do the work.**

Our mission statement guides how we work, collaborate, and grow as a team.

MISSION STATEMENT

> We **inspire** innovation in robotics, empowering our team and community through STEM education and collaboration, fostering future leaders in technology.

- **Everyone** has a **role** on our team
- Junior members are younger teammates who work on non-competition robots while being **mentored** by senior members
- Notice how we have no designated outreach role? **Every member** participates in **outreach** and **connecting** to our community!
- We **collaborate** across all roles, working as one to **improve** our robot, **strengthen** our team, and become the **best versions** of ourselves!

CONNECT : We explore ways to achieve our goals and engage with the local STEM community!

TEAM GOALS AND STEPS TAKEN TO ACHIEVE THEM:

Engage in STEM & FIRST outreach to over 1500 people in our community

- **Hosted** open houses and community events
- **Promoted** our team through school and community channels
- **Connected** with teams and grew our social media presence

Our Instagram, where we **connect** with other teams, update and **promote** our sponsors, and post our **achievements**!

Connect and learn from STEM professionals

- **Meeting** with past **SRT alumni** who moved on to professional fields
- **Connected** with **STEM professionals,** including **engineers** from **HOCS Projects** and **Tridyne**
- Applied our lessons from our school teachers to robotics

Many of our SRT alumni stay **involved with FIRST**, returning to visit, **mentoring** us, and **volunteering** at competitions!

Give back to FIRST

- We give back by regularly **volunteering** at FIRST events
- Assist with setup, support teams in the pit, reset the field, and help matches run smoothly

Our Grade 9 members **volunteered** at Clay Pot Meet #2, demonstrating **Gracious Professionalism**

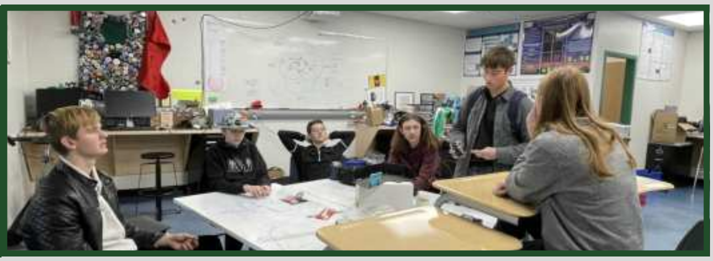
Achieve Worlds, or a World-Level Event

As a team, we have decided to push our skills to improve and become the best versions of ourselves. By excelling beyond our previous performances at competition, we are able to track our progress in improvement.

TEAM MEETINGS

We also hold **regular SRT team meetings** that include our team (14023), sister teams, and our junior members. In these meetings, we:

- Involve everyone in **robot development**, **sustainability**, **fundraising**, and **outreach**
- Discuss our **progress** in our goals and team plans
- **Debrief competitions** as a team, discussing successes and improvements



COLLABORATING AND CONNECTING WITH OTHER TEAMS

We **learn** from and **support** other teams through our participation in:

- Scrimmages - Online Platforms - Social Media - Local Meetups - Competition Pits -

And more to come! Some scrimmages we attended include:

Amazon Day Scrimmage @ TELUS Spark

Why the red jackets? Our team members forgot their green ones!

In this photo, our drive team is teaching a young kid how to drive the robot! Inspiring!

Canmore Trailblazer Bot Bash

In this event, our junior members were able to show off their robots! They got first-hand competition experience without the usual stakes.

Nelson Mandela Scrimmage

In this event, we were awarded the “Spirit & Gracious Professional Award” Banner!

Online Resources

This Engineering Portfolio was assembled with the help of **Team 16091** (Team Without a Cool Acronym) ‘s **Portfolio Guide**, as well as **FTC Mentor Brogan M. Pratt’s YouTube video guide**.

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IMPACT AND OUTREACH DETAILS

OPEN HOUSE

During our school’s **Grade 8 Open House**, our robotics team showcased the robot, our lab, and our achievements!

- Encouraged younger students to get **interested** in robotics and STEM
- Talked about our team, competitions, and experiences
- Presented a slideshow highlighting the **ideals and values of FIRST**

INSPIRING FUTURE GENERATIONS

Our team annually takes the opportunity to visit **Elbow Elementary School** to inspire the younger students.

- Taught them the **basics of coding, creating, innovating, and CAD**
- Hands-on exposure to STEM concepts
- **Encouraged** curiosity, creativity, and a lasting interest in engineering and technology.

STEAMOJI MENTORING

Team members **mentored students at Steamoji** in many different branches of STEM

- A large focus on coding and CAD
- Exposed students ages 4–15 to STEM concepts and hands-on learning
- Connected with over 200 apprentices through mentoring and outreach initiatives

REACH: Our efforts to inspire others to embrace the FIRST culture, while introducing them to the exciting world of STEM.

Outreach is essential to **promoting FIRST** and **inspiring** the next generation of innovators. By **connecting** with schools, local businesses, and our community, SRT highlights opportunities in **STEM** while **building support and leadership** within our team.

The table below shows events we hosted or attended to **introduce others to STEM:** total hours are calculated as the **sum of each member's individual contributions** (5 Members volunteering for 2 hours → 10 hours total). The table shows major events, total reach, including smaller demos exceeds 1500 people.

Date	Event	Hours	Impact
Jan 2025	Supporting Coding week at Elbow Valley Elementary School	54	200 people
Feb 2025	SHAD Ocean Presentation	80	40 people
March 2025	Family and Friends Event at SCHS	40	150 people
November 2025	Telus Spark Scrimmage	16	50 people
Jan 2026	Supporting Coding week at Elbow Valley Elementary School	54	200 people
March 2026	Grade 8 Open House at SCHS	50	250 people
March 2026	Canmore Trailblazer Bot Bash Scrimmage	48	100 people
April 2026	Calgary Wild FC with Future Leaders Foundation	10	120 People
April 2026	Steamoji Mentoring	96	250 people
And much more to come in the future!			

SUSTAINABILITY:

Our team's long-term success relies on sustainable funding and financial support.

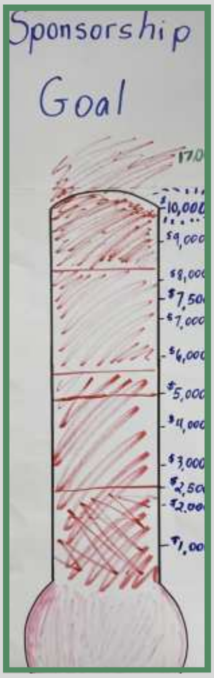
FUNDRAISING EFFORTS:

10/2025 - 12/2025
Purdy's Holiday Fundraiser
+$2366.44

1/2026 - 3/2026
Springbank 50/50 Raffle
+$9668.78

4/29/2026
Team Bottle drive
+$816.38

...AND MANY MORE!



FINANICAL SUPPORT

Financial support is **essential** to our team's success. It:

- Covers **costs** of building robots and attending competitions
- Allows us to focus on technical and strategic **development**
- Supports our ability to **compete, innovate, and grow**
- Plays a vital role in both **current success** and **future goals**

SPONSOR SUPPORT

We are incredibly **grateful** to all of our **sponsors** for their contributions to our team. In return for their financial support and mentorship, we:

- **Invite** sponsors to visit our classroom and competitions
- **Visit** sponsor workplaces to learn from their **processes** and apply those **insights** to our designs
- **Promote** sponsors on our robot, banners, social media, and team materials

We also continuously **explore new ways** to give back and show our appreciation!

HOCS PROJECTS · Tridyne PROJECTS · OPTIMA · ROCKY VIEW SCHOOLS · BONNETTS Energy Corp. · FIDELITY MANUFACTURING · trotec · SPOOL3D · HNW FINANCIAL · MNP

LONG-TERM FINANCIAL SUSTAINABILITY

To ensure **our program continues** beyond this year, we focus on saving wherever possible. We do this by:

- **Reusing** old robot parts, team uniforms, laser-cut panels, and trinkets
- Avoiding major robot changes in the second half of the season to **reduce costs**
- Continuing to **seek and secure sponsors** even after the robotics season ends

HOW DO WE BRING IN NEW MEMBERS YEAR AFTER YEAR?

Junior members are usually introduced through the school robotics class.

- Learn the basics of building, designing, CAD, and coding of a robot
- Learn what FTC is and how it works

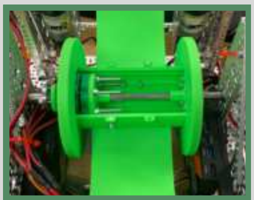
Based on their natural interests, senior members (as a student-led team) teach junior members more specialized roles

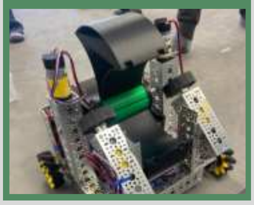
- Building and designing robots, using Fusion360, GoBuilda parts, etc
- Moving from block code to OnBotJava and RoadRunner/Auto
- Developing entrepreneurial skills (fundraising and sponsor outreach)
- And a great mix of many other roles!

JUNIOR ROBOTS

As a student-led team, **senior members mentor juniors** through hands-on work with **non-competitive robots built from spare parts**. This **low-pressure** setup helps them learn and experiment without risk. We maintain a team Google Drive with how-to-CAD videos and code examples to ensure knowledge is passed down each year.

- For example, younger members designed a **lift system** using REV bars and dual Viper Slides to evenly distribute weight and achieve high height.





- They also created a **3D-printed pinwheel intake** that funneled artifacts into a rotating sorter.
- When paired with computer vision, it has the **potential** to automatically organize pieces for maximum pattern points.

TEAM BONDING

This team is more than just robotics, **it's a family**. Our connections extend far **beyond the robotics classroom.** Through this program, we've built **lifelong friendships** and learned **valuable collaboration skills.**
Even after graduation, many alumni still come back to mentor us and spend time together. Robotics inspires us not only to **grow as students, but also as people.**

ADAPTING TO THE TEACHER'S STRIKE

Due to the Alberta teachers' strike, our team temporarily moved to a member's house to continue work on the FTC robot.

- This quick change kept us **on track** with design, build, and programming.
- The obstacle **highlighted our ability** to stay focused despite the challenges, strengthening our **commitment** to **perseverance** and **teamwork**, values that continue to **inspire** us throughout the season.

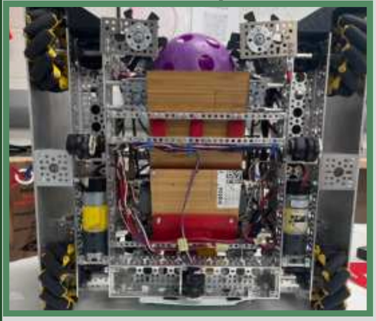
DESIGN:

Meet the robots of our team, explore our robot iterations and discover our unique build ideas.

MEET MADDISON, 14023'S DECODE ROBOT

Maddison, named in honor of an SRT alumnus, features **an active intake** with a **dual conveyor system** to efficiently collect artifacts and deliver them to the **flywheels** for consistent scoring.

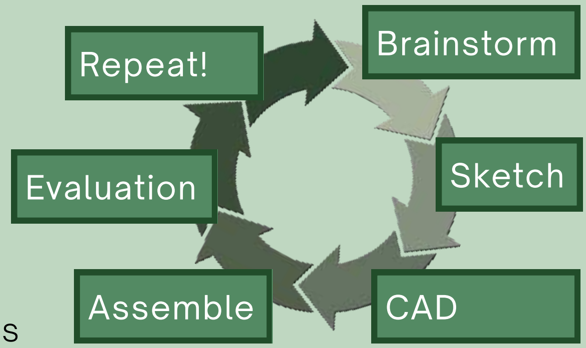
Our robot also includes an **adjustable flywheel launch system** and **mecanum wheels**, allowing for quick maneuvers and versatile strategies. Given the high-contact nature of this season's games, we designed **rigid aluminum side panels** that can tank impacts, allowing our drivers to stay focused on scoring.



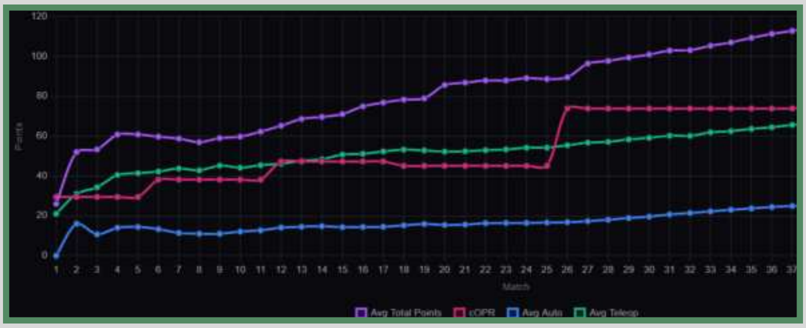
DRIVETRAIN	INTAKE	CONVEYOR	FLYWHEELS
• 435 RPM motors • Mecanum strafe wheels from goBuilda, allowing us to achieve holonomic movement • Three dead wheels for localization in autonomous	• Two 1620 RPM motors • GoBuilda boot wheels, intaking artifacts with speed • TPU wheel bumpers that protect wheels and funnels artifacts	• Four melon-robotics servos in total, two on each side • Symmetrical dual conveyor with GoBuilda timing belts to move artifacts from the intake to the flywheels	• Two 6000 RPM motors • Attached to a servo for adjustable angles, enabling shots from anywhere on the field • Rigid Rhinos wheels

SEASON OUTLINE AND DESIGN PROCESS

Our design process during the "**Design Modification**" sections outlined above follows six steps: **brainstorming**, **sketching**, **CAD modeling** in Fusion 360, **assembly**, **evaluation, and repeat**. We document every iteration in this portfolio, using performance data to drive our design cycle.



MATCH PERFORMANCE GRAPH



This **FTCSecrets graph (by Team 31000)** maps our seasonal growth across 4 events (37 total matches). It tracks Total Points (purple), Autonomous (blue), Teleop (green), and cOPR (red), visually demonstrating our **iterative performance improvements**.

MAJOR ROBOT ITERATIONS OVER THE SEASON

V1 (Name: Meghan) Meets 1 & 2

- Focused on finishing the robot quickly and **learning the game.**
- Simplicity and speed were prioritized over reliability
- **Pros**: Finished in time, simple design, helped us learn the game
- **Cons**: Shooting accuracy was ~40% due to wheel compression, and artifacts got stuck in the conveyor

V2 (Name: Maddison V) Meet 3 - Qualifiers

- Redesigned the robot with a **square chassis** and several upgrades to improve accuracy and autonomous.
- **Pros**: Better accuracy, improved autonomous, added dual conveyor and anti-jam features

- **Cons:** Still got pushed around, the lift was ineffective, artifacts still got stuck

V3 (Name: Maddison V) Meet 4 & 5 - Provincals & Worlds Premier Event

- Refined the robot by improving defense, consistency, and driver performance
- Removed the lift & switched conveyor belt servos
- **Pros**: More reliable, stronger defense, faster shooting, improved driver practice, and autonomous

STRATEGY/SCOUTING TEAM

We have a dedicated and hardworking **strategy/scouting team** that plays an important role in our **success**. They:

- **Analyze other teams**' strengths, weaknesses, and strategies to support **match planning**
- **Communicate** with teams to evaluate **alliance opportunities** and **build connections**
- Track robot performance and match data for **strategic decisions**
- **Coordinate** with judges and **collaborate** with other scouting teams at the team station
- **Analyze our gameplay** and provide feedback on what we can improve and learn from our mistakes.

INNOVATE:

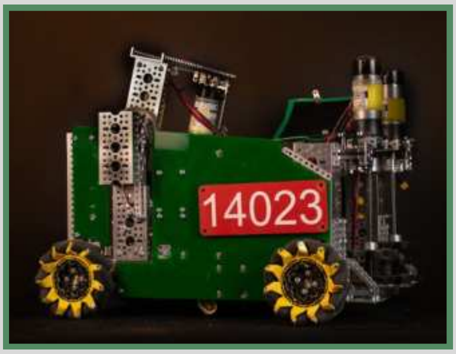
Meet our prototype designs, experimented ideas and robot evolution throughout the season!

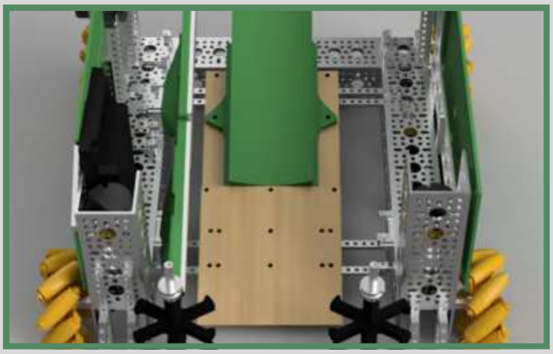
ROBOT EVOLUTION V1 - CLAYPOT MEET TWO & ARROWHEAD MEET TWO (#1 & #2)

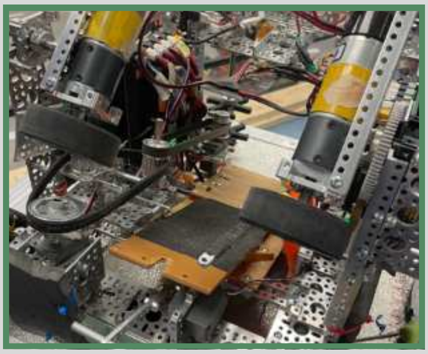
MEGHAN, OUR FIRST ROBOT ITERATION

Our initial robot design was named **Meghan**, after a fellow alumnus. Meghan featured:

- **One-sided conveyor belt system** that moved artifacts from the intake to the shooters
- Dual vertical intake with goBuilda bootwheels
- A set of **adjustable gecko-wheel shooters**
- **A small and compact robot**, which also enabled quick maneuvers and greater adaptability on the field







V1 - REFLECTION

Main purpose of Iteration 1 (Meghan)

- Get a functional robot ready to play (**Lost time** due to the teacher strike)
- **Testing** out basic ideas

Lessons for next time:

- High contact during gameplay required **added protection** for components
- The gecko wheels would **compress**, causing **inaccuracies** with shots
- A single conveyor belt caused artifacts to enter the flywheels **unevenly**, leading to **shots veering in one direction**
- It was too **difficult** for our drivers to **stop the belts in time** to prevent the artifacts from **firing early**

RETIRING TO A BACKUP ROBOT

After competing with Meghan, we developed a new robot chassis **based on a design that had worked well for us before**, resulting in our current robot, **Maddison**.

As a result, Meghan has been relegated to a backup role, allowing us to:

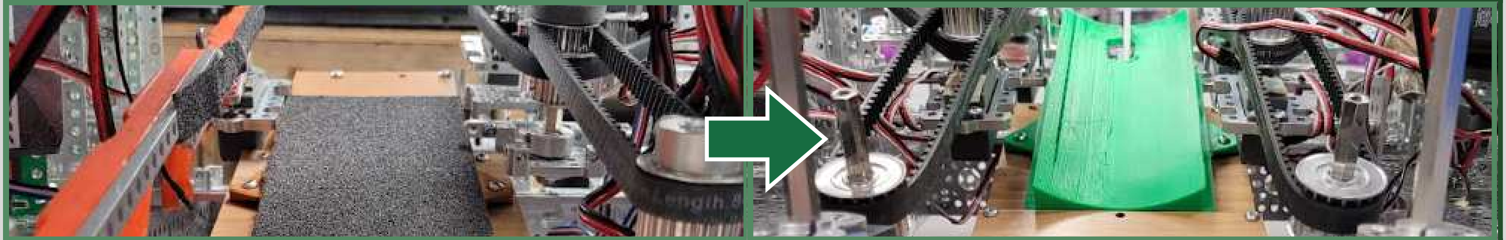
- Fine-tune code
- **Experiment** and **prototype** new ideas
- Help develop **junior members**' technical skills and driving abilities

All without **interrupting** work on our new robot! Thanks to Meghan, the future of our program is looking bright.

ROBOT EVOLUTION V2 - NAA'MIIS'STII INTERLEAGUE TOURNAMENT (#3)

This page highlights the key improvements made from Meghan's to Maddison's design, our second robot.

CONVEYOR BELT CHANGES



The robot uses a **3D-printed angled ramp** to **move artifacts** from the intake to the shooter via a conveyor. Our first design had a **single-sided belt**, which caused artifacts to **slip back** down and spin.

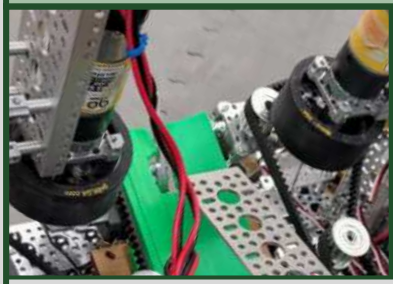
We **improved** it by using wider goBILDA timing belts and **adding a second belt on the opposite side**, gripping artifacts from both sides for **reduced slippage** and a more consistent feed into the shooter.

CHASSIS

Maddison has a **wider U-channel**, allowing for **easier maintenance access**. This improves on Meghan's design, where wiring and adjustments were more **tedious** and **difficult to access**.

Maddison also features a **symmetrical chassis**, which removes the need to account for the **offset in dead wheels**. That enables a **more precise and accurate autonomous performance** and pathing.

FLY WHEELS SWAP



Our **Gecko Flywheels** caused our shots to **drift left and right** due to them **expanding** and **compressing** while spinning.

We switched to **Rhino wheels**, which **do not** expand or compress. This improved our shooting consistency.

ARTIFACT BLOCKER

To prevent artifacts from contacting the flywheels **early**, we added a **servo with a block** at the end of the ramp.

This allows our drivers to focus on driving and reducing potential penalties.

V2 - REFLECTION

__Improvements of Iteration 2 (Maddison)__

- **Increased accuracy** by adding rhino wheels and double conveyor belts
- **Protected our wheels** with acrylic side panels
- Added a **lift** for endgame points
- Added an artifact blocker to help drivers **prevent premature firing**

__Lessons for next time:__

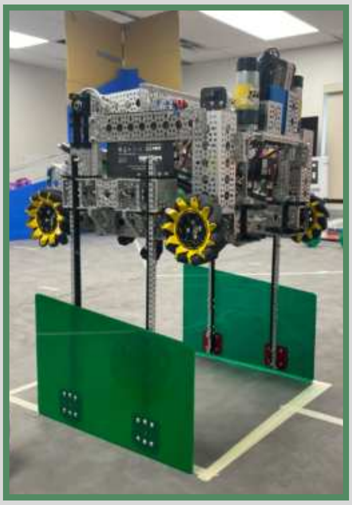
- Acrylic side panels were **not durable** enough for the high-contact gameplay
- The lift system was very **unstable** and was rarely high enough to accommodate alliance partners
- During autonomous, the robot **struggled to intake artifacts** from the field

ROBOT EVOLUTION V3 - ALBERTA CHAMPIONSHIP (#4)

TRADE-OFF: WEIGHT AND RISK VS. ENDGAME POINTS

In V2, we added a **rack-and-pinion lift** (driven by four servos) to allow teammates to park underneath for endgame points. During testing and competition, we identified several issues with the mechanism:

- The lift added weight, **reducing autonomous speed** and **overall consistency**
- Created a **risk**, as falling from 18 inches would **negatively impact** the **robot's performance**
- Most robots **could not fit underneath**, so the mechanism was **rarely used** and **not very effective**



Due to its **limited usefulness** and **negative impact** on performance, we **removed** it for provincials (V3).

WHEEL BUMPER/FUNNEL AND TPU INTAKE

Throughout our matches, we realized we were struggling to **intake artifacts stuck in the corners** of the field. To address this, we:

- Designed a **unique 3D-printed wheel bumper** that also **funnels artifacts** directly into our intake

- This allows us to **reach into corners** we previously couldn't, while **protecting** our wheels from damage.

ALUMINUM CUSTOM SIDE PANELS

Using a **CNC plasma cutter**, we designed and manufactured **custom aluminum side panels**.

Choosing metal instead of lighter materials **intentionally increased** the robot's weight, allowing **more resistance** to **defensive contact** and helping maintain **stable positioning** for **consistent performance.**

V3 - REFLECTION

Improvements of Iteration 3 (Maddison)

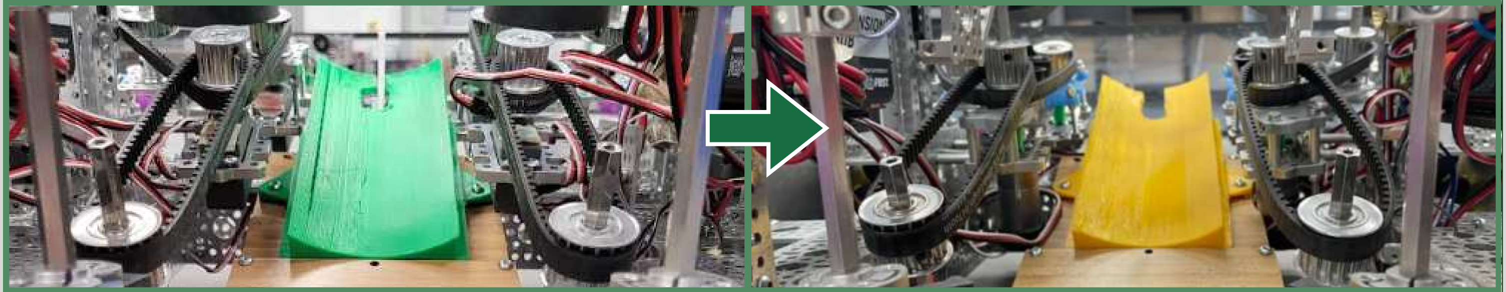
- **Removed the lift** to increase speed and stability
- Added metal side panels for **better protection** in **high-impact gameplay**
- Tested new intake wheels for autonomous intake, but they showed **little improvement** over the original boot wheels
- Added a funnel system with bumpers, which **better guided artifacts** into the intake and **protected** the wheels

Lessons for Next Time

- Lack of speed **hurt performance** in both autonomous and teleOp
- Improvements are not always **major redesigns**, as **refining smaller details** can be more effective

ROBOT EVOLUTION V3 - WESTERN EDGE PREMIER EVENT (#5)

CONVEYOR BELT CHANGES



Our final conveyor system upgrade integrated the Super Servo Plus from Melonbotics.

- Operating at 1000 RPM, the servos **outperform** goBILDA's Super Speed Servos by nearly four times, **reducing** cycle times by 4–6 seconds
- Improved **overall scoring efficiency** and robot output during matches.

However, the upgrade became one of the robot's **most technically demanding modifications**.

- Integration required **custom** aluminum brackets, **3D-printed components,** and **redesigned belt-tensioning systems**.
- Due to its **complexity**, our team has **repeated meetings** and **evaluated** whether the performance improvements justified the added **engineering complexity** and **build time.**

DRIVER PRACTICE AND MOCK MATCHES

In our group debrief, we identified a short amount of **driver practice time** as a key weakness. To address this, we **increased** practice sessions and **organized mock matches with junior robots** and our former robot **Meghan**.

This helped our drivers refine their skills while giving juniors a chance to showcase the robots they worked hard on.

V3 - REFLECTION (CONTINUED)

<u>**Improvements of Iteration 4**</u>

- Added faster servos, which **significantly increased shooting and conveyor speed**
- **Improved** overall cycle times and scoring **efficiency**
- **Increasing** driver practice time greatly **increased our scores**

<u>**Lessons Learned**</u>

- **Faster** and **more efficient** mechanisms can greatly **improve** match performance
- Driver practice is **just as important** as robot design and programming
- Focusing on both mechanical **improvements** and team **coordination** leads to more **consistent results**

CONTROL:

We use software and hardware in various ways to improve our robots functionality during matches!

AUTONOMOUS CODE

Our autonomous was designed for consistent and accurate movement using **Road Runner** and **three dead wheels**.

To ensure adaptability, we **coordinated** with alliance partners and **adjusted our routines** when needed, such as reducing our 9-artifact autonomous to 6 artifacts to **prevent collisions**.

Pathing: Drive to spike mark, → Collect Artifacts, → Shoot in alliance goals, -> Repeat

ODOMETRY PODS

We use **dead wheels** with **rotary encoders** to accurately track robot movement and position during autonomous.

Compared to mecanum wheel encoders, dead wheels experience **far less slip**, **improving accuracy** during high acceleration.

This allows us to **fine-tune** our autonomous routines for more **consistent** performance.

PROBLEMS & PROGRAM SOLUTIONS

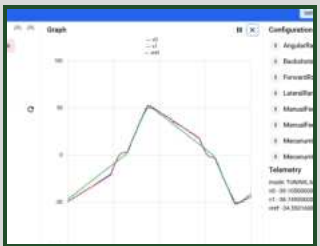
HEAVY ROBOT WEIGHT

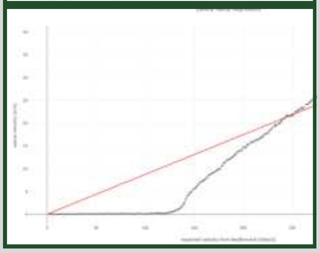
Problem: Our robot was **significantly heavier** due to the weight of the metal side panels. Our design **added defense**, which made movement **too slow** for an autonomous that can pick up all 12 artifacts.

Solution:

- Increased RoadRunner **acceleration limits**.
- Tuned kV, kA, and kS values through testing (and a few blown battery fuses!) (**Tuning graphs are to the right** →)

Now: Our robot moves **much faster** during autonomous **without losing stability**, allowing us to achieve our goal of a 12-artifact autonomous!





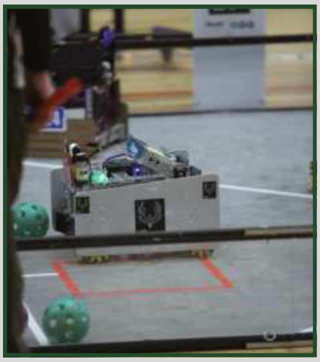
ARTIFACT COLLECTION CONSISTENCY

Problem: Different trajectory accelerations caused **inconsistent artifact collection**, as they would be knocked away without being intaken

Solution:

- Added **velocity constraints** to pathing and each trajectory, which **slowed and stabilized** the robot before intake

Now: Increased our autonomous intake success rate from 70% to 95%.



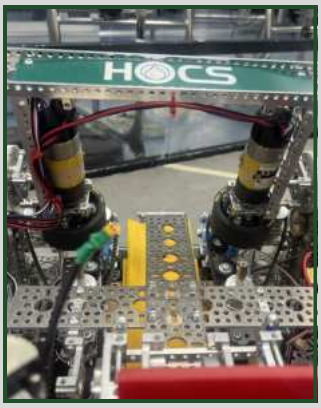
BATTERY-DEPENDENT FLYWHEELS

Problem: Flywheel speed dropped as battery voltage decreased, **reducing shot consistency** and requiring a **specific** starting voltage (13.55 V), which was **inconvenient** and caused **teleop power issues**.

Solution:

- Used **encoder-based velocity** control to maintain constant RPM

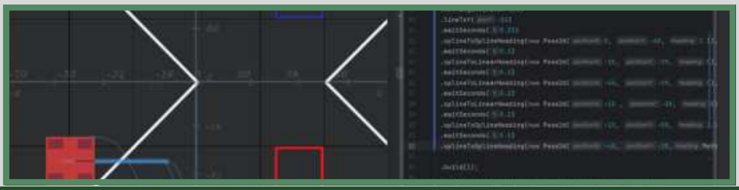
Now: Shooting is now consistent and accurate throughout autonomous, and fresh batteries can be used without running them down.



TESTING AUTONOMOUS CODE LIBRARIES

ROADRUNNER

- Road Runner is a **coordinate-based** motion library developed by ACME Robotics.
- It enables our robot to execute complex, high-velocity **trajectories** with advanced path-following
- Enables precise and reliable trajectory-based movement.



PEDROPATHING

- PedroPathing is a **Reactive Vector Follower** by Team 10158.
- Adapts to obstacles **in real time**.
- Extremely voltage dependent
- Despite being harder to set up, PedroPathing is **faster** and **more accurate** than Road Runner

LIBRARY COMPARISON AND TRADE-OFF ANALYSIS

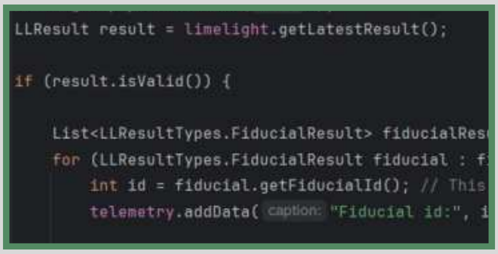
We have been working with **both Java Libraries**, comparing and contrasting their strengths and limitations.

PedroPathing **adjusts in real time** during a path, making it **responsive and precise**, but it is highly **battery-dependent** and can be volatile. In contrast, RoadRunner only **adjusts after autonomous**, reducing **real-time accuracy**. However, it remains **reliable** under **voltage drops**, an important **advantage** this year as we run multiple motors at full power

LIMELIGHT SMART CAMERA TESTING

Our team has also been experimenting with **Limelight Smart Cameras**, which are specialized for robotics and April Tag detection. We have:

- **Successfully detected** the **Obelisk AprilTag** to select accurate autonomous trajectories.
- **Tested** camera integration in a non-competitive environment to **improve reliability and precision.**
- Began **learning OpenCV** for future real-time vision processing and machine learning applications



```
LLResult result = limelight.getLatestResult();

if (result.isValid()) {

    List<LLResultTypes.FiducialResult> fiducialRes
    for (LLResultTypes.FiducialResult fiducial : f
        int id = fiducial.getFiducialId(); // This
        telemetry.addData( caption: "Fiducial id:", i
```

CAMERA INTEGRATION REFLECTION

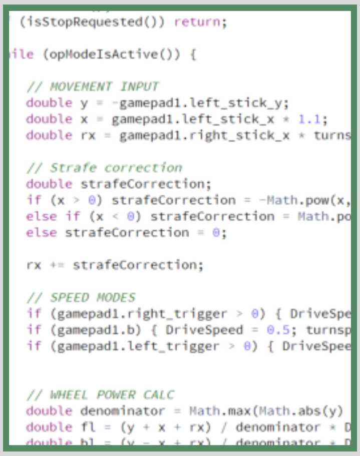
Our Limelight testing was **successful**, but full integration was **not practical** within our season timeline. While the camera **consistently detected AprilTags** and supported accurate path selection, implementing and tuning the system required **more time than available**.

This experience taught us the importance of **balancing innovation with reliability**. We plan to continue developing vision systems with Limelight and OpenCV for **future FTC seasons**.

TELE-OP DRIVE CODE

Our Tele-Op code was built using **OnBot Java** in the REV Hardware Client. It features:

- **Robot-centric control** enabled by compact robot design for better stability and movement
- **Preset motor positions** allow consistent **shooting angles** from different field locations
- **Adjustable shooting power** helps account for defense and positioning without overshooting
- **Multiple drive speeds** improve **accuracy** during endgame parking



```
 (isStopRequested()) return;

ile (opModeIsActive()) {

  // MOVEMENT INPUT
  double y = -gamepad1.left_stick_y;
  double x = gamepad1.left_stick_x * 1.1;
  double rx = gamepad1.right_stick_x * turns

  // Strafe correction
  double strafeCorrection;
  if (x > 0) strafeCorrection = -Math.pow(x,
  else if (x < 0) strafeCorrection = Math.po
  else strafeCorrection = 0;

  rx += strafeCorrection;

  // SPEED MODES
  if (gamepad1.right_trigger > 0) { DriveSpe
  if (gamepad1.b) { DriveSpeed = 0.5; turnsp
  if (gamepad1.left_trigger > 0) { DriveSpee


  // WHEEL POWER CALC
  double denominator = Math.max(Math.abs(y)
  double fl = (y + x + rx) / denominator * D
```

DRIVER CONTROLS

Driver 1

Driver 1 controls **robot movement** (therefore, the **aiming angles** toward the Alliance Goals)

Driver 2 controls the **shooter angle & power**, **intake**, **conveyor**, and **artifact blocker**.

Driver 2

DRIFT CORRECTION

Problem: Our robot **drifted** during tele-op due to **uneven weight distribution** and mechanical imbalance.

Solution:

- Programmed a **strafe correction feature** into the drive code.
- Automatically **compensated** for uneven motor outputs.

Now: Straighter and smoother driving, with improved driver control and navigation **accuracy**

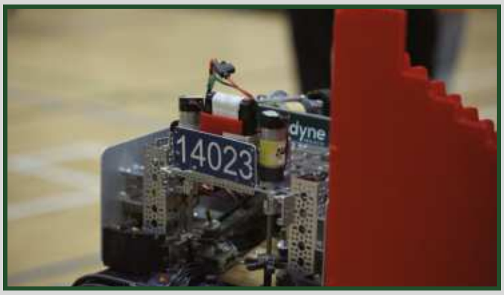
CONTROLLER BUTTON LIMITS

Problem: We ran out of controller buttons, making controls **confusing** during tele-op and **overly complicated**

Solution:

- Added a **toggle button** for the conveyor belts and intake motors.
- Used a **Boolean state** to switch the intake system on and off with one button.

Now: Simpler and more efficient driver controls, allowing for faster intake operation during matches.



THANK YOU FOR REVIEWING OUR PORTFOLIO, AND FOR YOUR SUPPORT! - SRT PHOENIX RESURRECTION, 14023